Hints: Functional sequences, series and spaces – 5. Hilbert spaces and orthogonal systems

Class work

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Expand a square. With $c_n = \langle x, e_n\rangle$ one gets $\left\|x - \sum_{n\leqslant N} c_n e_n\right\|^2 = \|x\|^2 - \sum_{n\leqslant N}|c_n|^2 \geqslant 0$, which is Bessel. More generally, for $y \in L$ and $x - Px \perp L$: $\|x-y\|^2 = \|x-Px\|^2 + \|Px-y\|^2$ (Pythagoras), so Px is the unique best approximation. Conversely, if Px is best, then $t \mapsto \|x - Px - tv\|^2$ is minimal at $t = 0$ for every $v \in L$, which gives $\langle x - Px, v\rangle = 0$.

Hint 2. Parseval $\Leftrightarrow$ completeness. By the identity in Hint 1, Parseval for x means exactly that the partial sums $S_N x = \sum_{n\leqslant N} c_n e_n$ tend to x, i.e. x lies in the closed span. If the system is complete, x is a limit of finite combinations, and since $S_N x$ is the best approximation from $\operatorname{span}(e_1, \dots, e_N)$ the distance $\|x - S_N x\|$ goes to 0.

Problem 2

Hint 1. It suffices to approximate every element of a dense class in the L^2 norm. Continuous functions are dense in $L^2[0, 2\pi]$, and a continuous function can be changed on a short interval near 2π to become periodic at small L^2 cost. A periodic continuous function is a uniform limit of trigonometric polynomials (Fejér, or Stone–Weierstrass), and $\|g\|_2 \leqslant \sqrt{2\pi}\,\|g\|_\infty$.

Hint 2. Parseval for real continuous periodic f: $\frac{1}{\pi}\int_0^{2\pi} f^2 = \frac{a_0^2}{2} + \sum_{n\geqslant 1}(a_n^2 + b_n^2)$. If a continuous f is orthogonal to all e^{inx}, all its Fourier coefficients vanish, so $\int |f|^2 = 0$ by Parseval, and a continuous function with $\int |f|^2 = 0$ is identically 0.

Problem 3

Hint 1. Start in $L^2[0, 1]$, where $\langle x^a, x^b\rangle = \frac{1}{a+b+1}$. The squared distance from x^m to $\operatorname{span}(x^{\lambda_1}, \dots, x^{\lambda_N})$ is a ratio of two Gram determinants, and Gram determinants of this kind are Cauchy determinants $\det(\frac{1}{a_i + b_j})$, which have a product formula.

Hint 2. The result is $\operatorname{dist}_{L^2}(x^m, \operatorname{span}\{x^{\lambda_k}\}_{k\leqslant N})^2 = \frac{1}{2m+1}\prod_{k=1}^{N}\left(\frac{m-\lambda_k}{m+\lambda_k+1}\right)^2$. For m not among the exponents, the infinite product $\prod(1 - \frac{2m+1}{m+\lambda_k+1})$ tends to 0 iff $\sum \frac{1}{\lambda_k} = \infty$. So in L^2, all monomials (hence, by Weierstrass, everything) can be approximated exactly when the series diverges.

Hint 3. From L^2 to uniform: for $m \geqslant 1$, $x^m = m\int_0^x t^{m-1}dt$, and by Cauchy–Schwarz $\left|\int_0^x (t^{m-1} - q(t))\,dt\right| \leqslant \|t^{m-1} - q\|_{L^2}$. Approximating t^{m-1} in L^2 by combinations of $t^{n_k - 1}$ (their exponents still have divergent reciprocal sum) and integrating gives uniform approximation of x^m by combinations of the x^{n_k}; $1 = x^{n_0}$ is in the system. Conversely, if $\sum \frac{1}{n_k} < \infty$, some x^m has positive L^2 distance from the span, and $\|\cdot\|_2 \leqslant \|\cdot\|_\infty$ on $[0, 1]$.

Problem 4

Hint 1. Let $A_n = \frac{1}{n}\sum_{i=1}^{n} U^i$; then $\|A_n\| \leqslant 1$. On invariant vectors $A_n x = x$. On vectors of the form $x = Uy - y$ the sum telescopes: $A_n x = \frac{U^{n+1}y - Uy}{n} \to 0$.

Hint 2. Show $H = \ker(U - I) \oplus \overline{\mathrm{Ran}(U - I)}$: the orthogonal complement of the range is $\ker(U^* - I)$, and for unitary U, $U^*x = x \iff Ux = x$. The uniform bound $\|A_n\| \leqslant 1$ extends $A_n \to 0$ from the range to its closure, which gives $A_n x \to Px$.

Hint 3. The shift on $\ell^2(\mathbb{Z})$ is unitary, and its invariant vectors are constant sequences, which lie in ℓ^2 only if they are 0. Hence $P = 0$ and the averages tend to 0 in ℓ^2.

Problem 5

Hint 1. Gram–Schmidt gives P_n of degree exactly n, orthogonal to every polynomial of degree less than n. (Rodrigues' formula $P_n = c_n \frac{d^n}{dx^n}(x^2 - 1)^n$ gives the same polynomials: check orthogonality by integrating by parts.)

Hint 2. Three-term recurrence: expand xP_n (degree $n + 1$) in $P_0, \ldots, P_{n+1}$. The coefficient of P_k is proportional to $\langle xP_n, P_k\rangle = \langle P_n, xP_k\rangle$, which vanishes for $k + 1 < n$; so $xP_n = a_n P_{n+1} + b_n P_n + c_n P_{n-1}$, and $b_n = 0$ by parity.

Hint 3. Roots: if P_n changed sign in $(-1, 1)$ only at $r < n$ points $x_1, \ldots, x_r$, then $P_n(x) \prod(x - x_i)$ would have constant sign and yet integral 0 by orthogonality. So there are n simple roots in $(-1, 1)$. Interlacing: use the recurrence (or the Christoffel–Darboux identity, which gives $P'_{n+1}P_n - P_{n+1}P'_n > 0$ for a suitable normalization) to show that P_{n+1} alternates in sign at consecutive roots of P_n, then count the roots.